

Xilong Zhou

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Education

- Ph.D., Computer Science, Texas A&M University.** Aug 2018 – Aug 2024
Advisor: Dr. Ergun Akleman and Dr. Nima Kalantari,
Dissertation: Towards Practical and Robust Material Acquisition and Generation.
- M.S.[†], Visualization Department, Texas A&M University.** Aug 2016 – May 2018
Advisor: Dr. Ergun Akleman,
[†] Transfer to Computer Science Ph.D. Program.
- M.S., Petroleum Engineering, Texas A&M University.** Aug 2014 – Aug 2016
Advisor: Dr. Jenn-Tai Liang,
Thesis: Bulk and Adsorption Properties of Surfactant Entrapped in Polyelectrolyte Complex Nanoparticles.
- B.E., Petroleum Engineering, China University of Petroleum (Beijing).** Sep 2010 – Jun 2014

Research Interests

Generative AI, Video/3D Content/Material Generation and Editing, Neural Rendering, Novel View Synthesis, Inverse Rendering, and Relighting

Experiences

- Postdoc Researcher, Max Planck Institute for Informatics.** Sep 2024 – Present
Advisor: Dr. Christian Theobalt.
Topics: Generative AI, Video Editing, Neural Rendering, and Relighting.
- Research Intern, Adobe Research.** Feb – Jul 2024
Advisor: Dr. Miloš Hašan.
Topics: Realistic Diffusion-based Material Generation.
- Research Intern, Meta Reality Lab.** Sep – Dec 2022
Advisor: Dr. Chakravarty R. Alla Chaitanya.
Topics: Lightweight Novel View Synthesis.
- Research Intern, Adobe Research.** May – Aug 2022
Advisor: Dr. Miloš Hašan.
Topics: Realistic GAN-based Material Generation.
- Research Intern, Adobe Research.** May – Aug 2021
Advisor: Dr. Miloš Hašan.
Topics: Semi-procedural Material Priors & Editable Material Generation.

Publications

* Equal Contribution.

Preprints

- [1] **Xilong Zhou**, Jianchun Chen*, Pramod Rao*, Timo Teufel, Linjie Lyu, Tigran Minasian, Oleksandr Sotnychenko, Xiao-Xiao Long, Marc Habermann, and Christian Theobalt. OLatverse: A Large-scale Real-world Object Dataset with Precise Lighting Control. *arXiv preprint, Nov 2025.*
- [2] **Xilong Zhou***, Bao-Huy Nguyen*, Loïc Magne, Vladislav Golyanik, Thomas Leimkühler, and Christian Theobalt. Splat the Net: Radiance Fields with Splattable Neural Primitives. *arXiv preprint, Oct 2025.*
- [3] **Xilong Zhou**, Pedro Figueiredo, Miloš Hašan, Valentin Deschaintre, Paul Guerrero, Yiwei Hu, and Nima Khademi Kalantari. RealMat: Realistic Materials with Diffusion and Reinforcement Learning. *arXiv preprint,*

Sep 2025.

Journal/Conference Articles

- [4] Pramod Rao, **Xilong Zhou***, Abhimitra Meka*, Gereon Fox, Mallikarjun B R, Fangneng Zhan, Tim Weyrich, Bernd Bickel, Hanspeter Pfister, Wojciech Matusik, Thabo Beeler, Mohamed Elgharib, Marc Habermann, and Christian Theobalt. 3DPR: Single Image 3D Portrait Relighting with Generative Priors. *SIGGRAPH Asia 2025*.
- [5] Avinash Paliwal, **Xilong Zhou**, Andrii Tsarov, and Nima Kalantari. PanoDreamer: Optimization-Based Single Image to 360 3D Scene With Diffusion. *SIGGRAPH Asia 2025*.
- [6] Timo Teufel*, Pulkit Gera*, **Xilong Zhou**, Umar Iqbal, Pramod Rao, Jan Kautz, Vladislav Golyanik, and Christian Theobalt. HumanOLAT: A Large-Scale Dataset for Full-Body Human Relighting and Novel-View Synthesis. *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025.
- [7] Avinash Paliwal, **Xilong Zhou**, Wei Ye, Jinhui Xiong, Rakesh Ranjan, and Nima Kalantari. RI3D: Few-Shot Gaussian Splatting With Repair and Inpainting Diffusion Priors. *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025.
- [8] **Xilong Zhou**, Miloš Hašan, Valentin Deschaintre, Paul Guerrero, Yannick Hold-Geoffroy, Kalyan Sunkavalli, and Nima Khademi Kalantari. PhotoMat: A Material Generator Learned from Single Flash Photos. *SIGGRAPH 2023*.
- [9] **Xilong Zhou**, Miloš Hašan, Valentin Deschaintre, Paul Guerrero, Kalyan Sunkavalli, and Nima Khademi Kalantari. A Semi-Procedural Convolutional Material Prior. *Computer Graphics Forum (Eurographics 2023)*.
- [10] **Xilong Zhou**, Miloš Hašan, Valentin Deschaintre, Paul Guerrero, Kalyan Sunkavalli, and Nima Khademi Kalantari. TileGen: Tileable, Controllable Material Generation and Capture. *SIGGRAPH Asia 2022*.
- [11] **Xilong Zhou** and Nima Khademi Kalantari. Look-Ahead Training with Learned Reflectance Loss for Single-Image SVBRDF Estimation. *ACM Transactions on Graphics (SIGGRAPH Asia 2022)*.
- [12] **Xilong Zhou** and Nima Khademi Kalantari. Adversarial Single-Image SVBRDF Estimation with Hybrid Training. *Computer Graphics Forum (Eurographics 2021)*.
- [13] **Xilong Zhou**, Jenn-Tai Liang, Corbin D Andersen, Jiajia Cai, and Ying-Ying Lin. Enhanced Adsorption of Anionic Surfactants on Negatively Charged Quartz Sand Grains Treated with Cationic Polyelectrolyte Complex Nanoparticles. *Colloids and Surfaces A: Physicochemical and Engineering Aspects*, 553, 397-405, 2018.

Selected Ongoing Research

Physically Plausible Video Generation and Editing	<i>Project Leader</i>
◦ Developing a diffusion-based framework for physically plausible video editing ◦ Enables material editing, object manipulation, and lighting-aware adjustments	
Realistic Multi-View 3D Generation under Lighting Control	<i>Project Leader</i>
◦ Building a diffusion-based 3D generative model trained on real-world data ◦ Supports consistent view synthesis and controllable lighting	

Academic Service

Reviewer: SIGGRAPH, SIGGRAPH Asia, Eurographics, Pacific Graphics, ICCV, TOG, IEEE TVCG, IEEE TPAMI, CGF, Computers & Graphics.

Selected Awards

Travel Grant, CSCE Department, Texas A&M University.	<i>2023</i>
National First Prize of National Petroleum Engineering Design Competition.	<i>2013</i>
Honorable Mention of International Mathematical Contest in Modeling.	<i>2013</i>
National Second Prize of National Mathematics Modeling Contest.	<i>2012</i>

Media Coverage

SIGGRAPH Thesis Fast Forward , Ph.D. Dissertation.	2024
Two Minutes Paper , Paper [8].	2023

Teaching

Co-Instructor , Graphics Seminar, Saarland University.	<i>Summer 2025</i>
Teaching Assistant , Machine Learning, Texas A&M University.	<i>Fall 2020, Spring 2022</i>
Teaching Assistant , Analysis of Algorithms, Texas A&M University.	<i>Summer 2019, Fall 2021</i>
Teaching Assistant , Programming, Texas A&M University.	<i>Spring 2021</i>
Teaching Assistant , Data Structure and Algorithm, Texas A&M University.	<i>Spring 2019</i>
Teaching Assistant , Discrete Structure for Computing, Texas A&M University.	<i>Fall 2019, Fall 2018</i>
Teaching Assistant , Computer for Visualization, Texas A&M University.	<i>Spring/Summer 2017</i>
Teaching Assistant , Formation Evaluation, Texas A&M University.	<i>Spring 2016</i>
Teaching Assistant , Unconventional Oil and Gas, Texas A&M University.	<i>Fall 2015</i>